

Experiment 2

Title: Predicting Mental Health Outcomes from Social Media Usage

1. Dataset Source

- **Source Link:**

<https://www.kaggle.com/datasets/souvikahmed071/social-media-and-mental-health>

Source Description:

The dataset was collected to study correlations between social media usage patterns and general mental well-being. Mental health indicators are **self-reported** through survey questions covering distraction, validation seeking, social comparison, ADHD-like symptoms, anxiety, self-esteem, and depression.

2. Dataset Description

- **Number of Samples:** 481
- **Number of Features:** 21
- **Data Type:** Tabular CSV
- **Nature of Data:**
 - Anonymized survey responses
 - Mostly **categorical and ordinal** (Likert scale: 1–5)
 - Few missing values
 - No clinical diagnosis (purely self-reported mental health indicators)

Feature Categories

A. Demographic Features

- Timestamp
- Age (numeric)
- Gender (categorical)
- Relationship Status (categorical)
- Occupation Status (categorical)
- Organization Affiliation (categorical)

B. Social Media Usage Features

- Social media usage (Yes/No)
- Common platforms used (multi-select: Instagram, Facebook, Twitter, YouTube, TikTok, etc.)
- Average daily time spent on social media (categorical bins: <1 hr, 2–3 hrs, 4–5 hrs, etc.)

C. Behavioral & Mental Health Indicators (Likert Scale: 1–5)

- Distraction / ADHD-like behavior
- Anxiety symptoms
- Self-esteem levels
- Depression-related symptoms

(1 = Never / Not at all, 5 = Always / Very often)

Target Variables

“18. How often do you feel depressed or down?”

- Type: Continuous (1–5 Likert scale)
- Interpretation: Higher value indicates higher frequency of depressive feelings

3. Mathematical Formulation of the Algorithms**3.1 Multiple Linear Regression**

The linear regression model is defined as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

Objective Function (MSE Minimization):

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

3.2 Ridge Regression (L2 Regularization)

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

- Reduces coefficient magnitude
- Handles multicollinearity
- Does not eliminate features

3.3 Lasso Regression (L1 Regularization)

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- Performs feature selection
- Forces less important coefficients to zero

4. Algorithm Limitations

Common to All Models

- Assumes linear relationship between predictors and target
- Sensitive to outliers
- Performance limited by noisy survey data
- Cannot capture complex non-linear interactions

Multiple Linear Regression

- Highly sensitive to multicollinearity
- Can overfit when many correlated predictors exist

Ridge Regression

- Does not remove irrelevant features
- Requires careful tuning of regularization parameter

Lasso Regression

- Unstable with highly correlated features
- May discard useful predictors arbitrarily

5. Methodology / Workflow

Data Acquisition

- Download [smmh.csv](#) from Kaggle

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
1	Timestamp	1. What is 2. Gender 3. Reliator 4. Occupal 5. What ty 6. Do you 7. What sc 8. What is 9. How oft 10. How o 11. Do you 12. On a sc 13. On a sc 14. Do you 15. On a sc 16. Follow 17. How o 18. How o 19. On a sc 20. On a scale of 1 to 5, how often																						
2	4/18/2022	21	Male	In a relatic	University	University	Yes	Facebook, Between 2	5	3	2	5	2	5	2	5	2	3	2	5	4	5		
3	4/18/2022	21	Female	Single	University	University	Yes	Facebook, More than 4	3	2	4	5	4	5	1	1	1	5	4	5				
4	4/18/2022	21	Female	Single	University	University	Yes	Facebook, Between 3	3	2	1	2	5	4	3	3	1	4	2	5				
5	4/18/2022	21	Female	Single	University	University	Yes	Facebook, More than 4	2	1	3	5	3	5	1	2	4	3	2					
6	4/18/2022	21	Female	Single	University	University	Yes	Facebook, Between 2	3	5	4	4	5	5	3	3	3	4	4	1				
7	4/18/2022	22	Female	Single	University	University	Yes	Facebook, Between 2	4	4	2	3	4	3	4	4	4	3	2	4				
8	4/18/2022	21	Female	Married	University	University	Yes	Facebook, Between 3	4	3	2	2	4	3	5	3	4	5	5	3				
9	4/18/2022	21	Female	In a relatic	University	University	Yes	Facebook, More than 5	2	3	3	3	3	1	1	3	1	5	5	1				
10	4/18/2022	21	Female	In a relatic	University	University	Yes	Facebook, More than 5	2	3	3	3	1	1	1	3	1	5	5	1				
11	4/18/2022	20	Male	Single	University	University	No	Reddit, Pin Less than 1	1	1	1	1	1	1	1	1	1	1	1	1				
12	4/18/2022	24	Male	Single	University	University	Yes	Facebook, Between 2	5	5	2	4	3	5	4	2	3	5	4	5				
13	4/18/2022	21	Female	Single	University	University	Yes	Facebook, Between 3	4	4	2	5	3	4	4	2	2	3	3	4	2			
14	4/18/2022	22	Female	Single	University	University	Yes	Facebook, Between 3	3	4	2	4	4	2	4	2	1	3	1	5				
15	4/18/2022	21	Female	Single	University	University	Yes	Facebook, Between 3	5	3	2	1	4	4	2	3	2	3	3	3				
16	4/18/2022	22	Male	Single	University	University	Yes	Facebook, More than 5	5	5	5	5	5	5	5	5	5	5	5	5				
17	4/18/2022	23	Male	In a relatic	University	Private	Yes	Facebook, Between 1	3	4	2	3	4	4	3	2	2	3	4	2				
18	4/18/2022	21	Male	Single	University	University	Yes	Facebook, Less than 2	2	2	1	2	4	2	1	3	1	4	1	3				
19	4/18/2022	25	Male	Single	School Stu	N/A	Yes	Facebook, Between 4	5	5	4	4	3	3	1	4	2	2	2	2				
20	4/18/2022	28	Male	Single	University	University	Yes	Facebook, Between 1	3	4	2	4	5	5	2	3	1	3	3	1				
21	4/18/2022	34	Male	Divorced	Salaried W	Private	Yes	Facebook, Between 1	2	4	3	3	4	1	1	5	3	2	2	2				
22	4/18/2022	22	Male	In a relatic	University	School, Un	Yes	Facebook, Between 2	4	2	2	3	3	2	2	2	1	2	3	2				
23	4/18/2022	23	Male	Single	University	University	Yes	Facebook, More than 3	3	2	4	2	4	2	5	2	1	3	3	1				
24	4/18/2022	26	Female	Single	University	University	Yes	Facebook, Between 3	3	4	3	3	4	4	4	3	3	4	4	4				
25	4/18/2022	23	Female	Married	University	University	Yes	Facebook, Between 1	3	2	2	4	4	3	3	2	3	4	4	1				
26	4/18/2022	23	Male	Single	University	N/A	Yes	Facebook, Between 3	3	3	5	5	5	5	4	1	3	3	5	5				

Preprocessing

- Load dataset using pandas

```
[1] ✓ 2s
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LinearRegression, Lasso, Ridge
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

- Reading the Dataset Initial Values

```
df = pd.read_csv('smmh.csv')
print("Shape:", df.shape)
print("\nColumns:\n", df.columns.tolist())
df.head()

*** Shape: (481, 21)

Columns:
['Timestamp', '1. What is your age?', '2. Gender', '3. Relationship Status', '4. Occupation Status', '5. What type of organizations are you affiliated with?']
```

Timestamp	1. What is your age?	2. Gender	3. Relationship Status	4. Occupation Status	5. What type of organizations are you affiliated with?	6. Do you use social media?	7. What social media platforms do you commonly use?	8. What is the average time you spend on social media every day?	9. How often do you find yourself using Social media without a specific purpose?	10. ...	11. Do you feel restless if you haven't used Social media in a while?	12. On a scale of 1 to 5, how easily are you distracted by worries?	13. On a scale of 1 to 5, how much are you bothered by concentration on things?	14. Do you find it difficult to concentrate on things?	15. On a scale of 1-5, how often do you compare yourself to other successful people through the use of social media?	16. Following the previous question, how do you feel about these comparisons, generally speaking?	17. How often do you look to seek validation from features of social media?	18. How often do you feel depressed or down?	19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate?	20. On a scale of 1 to 5, how often do you face issues regarding sleep?	
0	4/18/2022 19:18:47	21.0	Male	In a relationship	University Student	University	Yes	Facebook, Twitter, Instagram, YouTube, Discord...	Between 2 and 3 hours	5	...	2	5	2	5	2	3	2	5	4	5
1	4/18/2022 19:19:28	21.0	Female	Single	University Student	University	Yes	Facebook, Twitter, Instagram, YouTube, Discord...	More than 5 hours	4	...	2	4	5	4	5	1	1	5	4	5
2	4/18/2022 19:25:59	21.0	Female	Single	University Student	University	Yes	Facebook, Instagram, YouTube, Pinterest	Between 3 and 4 hours	3	...	1	2	5	4	3	3	1	4	2	5
3	4/18/2022 19:29:43	21.0	Female	Single	University Student	University	Yes	Facebook, Instagram	More than 5 hours	4	...	1	3	5	3	5	1	2	4	3	2
4	4/18/2022 19:33:31	21.0	Female	Single	University Student	University	Yes	Facebook, Instagram, YouTube	Between 2 and 3 hours	3	...	4	4	5	5	3	3	3	4	4	1
5 rows x 21 columns																					

3. Clean column names

```
# Clean column names (remove extra spaces)
df.columns = [col.strip() for col in df.columns]
```

4. Defining Target Variable

```
# Target Variable (Single question - Simple & Clear)
target = '18. How often do you feel depressed or down?'
y = df[target].astype(float)
```

5. Encode categorical variables using one-hot encoding

```
# Preprocessor
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features),
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False), categorical_features)
    ])
```

6. Map time spent categories to numeric values

```
# Map Time Spent to Numeric
time_mapping = {
    'Less than an Hour': 0.5,
    'Between 1 and 2 hours': 1.5,
    'Between 2 and 3 hours': 2.5,
    'Between 3 and 4 hours': 3.5,
    'Between 4 and 5 hours': 4.5,
    'More than 5 hours': 6.0
}

X['time_spent_num'] = X['8. What is the average time you spend on social media every day?'].map(time_mapping)
X = X.drop(columns=['8. What is the average time you spend on social media every day?'])
```

```
# Identify numeric and categorical columns
numeric_features = ['1. What is your age?', 'time_spent_num'] + \
    [col for col in X.columns if 'scale of 1' in col]

categorical_features = [col for col in X.columns if col not in numeric_features]

print("Numeric features:", numeric_features)
print("Categorical features:", categorical_features)

Numeric features: ['1. What is your age?', 'time_spent_num', '12. On a scale of 1 to 5, how easily distracted are you?', '13. On a scale of 1 to 5, how
Categorical features: ['2. Gender', '3. Relationship Status', '4. Occupation Status', '9. How often do you find yourself using Social media without a sp
```

7. Apply feature scaling using StandardScaler

```
# Feature Selection
features = [
    '1. What is your age?',
    '2. Gender',
    '3. Relationship Status',
    '4. Occupation Status',
    '8. What is the average time you spend on social media every day?',
    '9. How often do you find yourself using Social media without a specific purpose?',
    '10. How often do you get distracted by Social media when you are busy doing something?',
    '12. On a scale of 1 to 5, how easily distracted are you?',
    '13. On a scale of 1 to 5, how much are you bothered by worries?',
    '15. On a scale of 1-5, how often do you compare yourself to other successful people through the use of social media?',
    '17. How often do you look to seek validation from features of social media?'
]

X = df[features].copy()
```

Exploratory Data Analysis

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

warnings.filterwarnings('ignore')

# -----
# 1. Load and prepare the data
# -----

df = pd.read_csv('smmh.csv')

# Clean column names (very important for this dataset)
df.columns = [col.strip() for col in df.columns]

print("Dataset shape:", df.shape)
print("\nColumns:\n", df.columns.tolist()[:15], "...") # truncated

# Convert Likert-scale columns to numeric (they should already be, but just in case)
likert_cols = [col for col in df.columns if 'scale of 1' in col or
    col.startswith(('9.', '10.', '11.', '12.', '13.', '14.', '15.', '16.', '17.)) or
    col in ['18. How often do you feel depressed or down?',
    '19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate?',
    '20. On a scale of 1 to 5, how often do you face issues regarding sleep?']]

for col in likert_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Age should be numeric
df['1. What is your age?'] = pd.to_numeric(df['1. What is your age?'], errors='coerce')
```

1. Distribution of age and mental health scores

```
#
# A. Distribution of Age
#

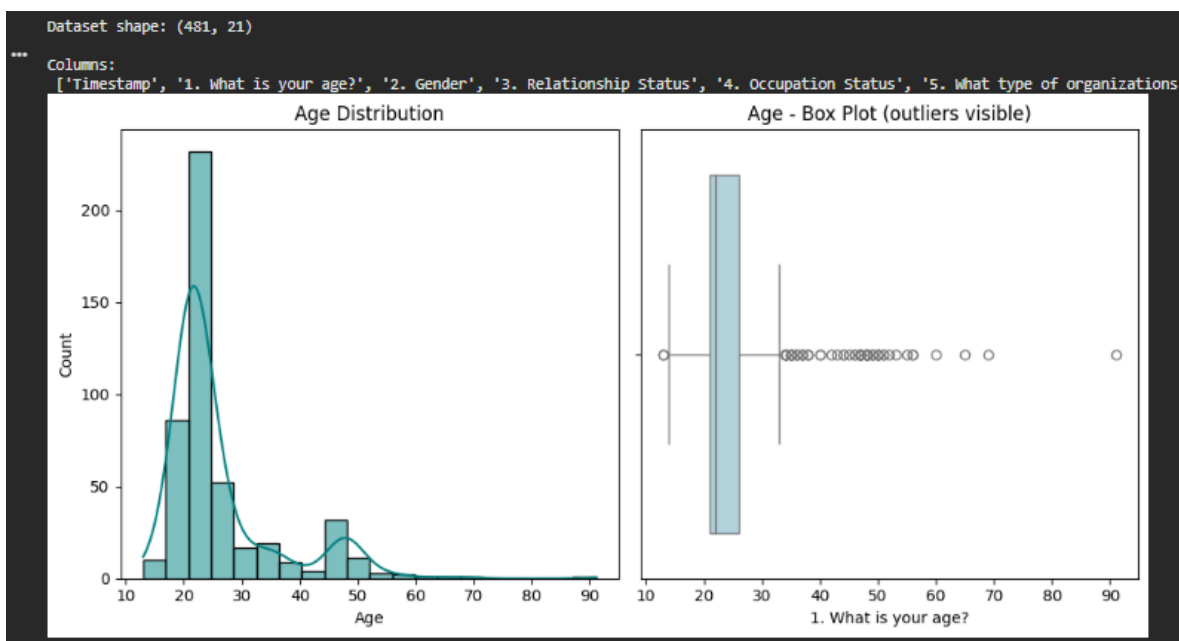
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
sns.histplot(df['1. What is your age?'], bins=20, kde=True, color='teal')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')

plt.subplot(1, 2, 2)
sns.boxplot(x=df['1. What is your age?'], color='lightblue')
plt.title('Age - Box Plot (outliers visible)')

plt.tight_layout()
plt.show()

# Basic stats
print("\nAge statistics:")
print(df['1. What is your age?'].describe())
```



1. Age Distribution – Histogram with Density Curve

Graph Type: Histogram (with KDE curve)

Description:

This graph shows the distribution of respondents' ages in the dataset.

- The bars represent the number of participants in each age group.
- The smooth curve (KDE) shows the overall density pattern.

- Most respondents are between **20–25 years**, indicating the dataset mainly represents young adults.
- The distribution is slightly right-skewed, with fewer older participants.

2. Age Distribution – Box Plot (Outliers Visible)

Graph Type: Box Plot

Description:

This graph summarizes the age distribution using quartiles.

- The box represents the **interquartile range (IQR)** (middle 50% of data).
- The line inside the box represents the **median age**.
- The whiskers show the range of typical values.
- Individual dots represent **outliers**, showing a few older participants (above 50–90 years).

This plot helps identify skewness and extreme age values in the dataset.

```
Age statistics:
count    481.00000
mean     26.13659
std       9.91511
min      13.00000
25%      21.00000
50%      22.00000
75%      26.00000
max      91.00000
Name: 1. What is your age?, dtype: float64
```

```
# B. Distribution of Mental Health Scores
# -----

# Define the main mental health outcome columns
mh_cols = [
    '18. How often do you feel depressed or down?',
    '19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate?',
    '20. On a scale of 1 to 5, how often do you face issues regarding sleep?'
]

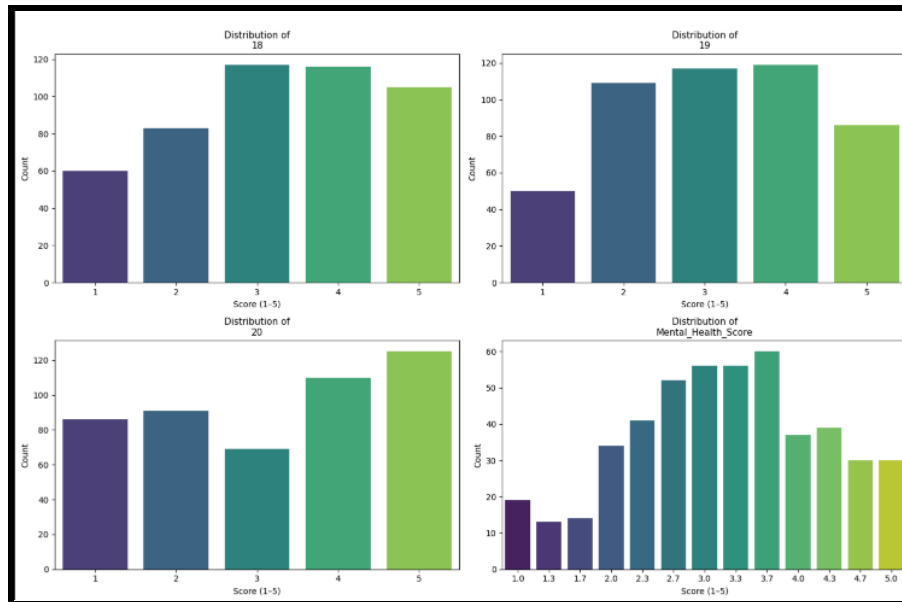
# Optional: create a simple composite score
df['Mental_Health_Score'] = df[mh_cols].mean(axis=1, skipna=True)

# Plot distributions
plt.figure(figsize=(15, 10))

for i, col in enumerate(mh_cols + ['Mental_Health_Score'], 1):
    plt.subplot(2, 2, i)
    sns.countplot(x=df[col].round(1), palette='viridis')
    plt.title(f'Distribution of {col.split(".")[0]}', fontsize=11)
    plt.xlabel('Score (1-5)')
    plt.ylabel('Count')

plt.suptitle('Distribution of Key Mental Health Indicators', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()

print("\nMental health items - descriptive statistics:")
print(df[mh_cols + ['Mental_Health_Score']].describe().round(2))
```

1. Distribution of Question 18 – Feeling Depressed or Down

Graph Type: Bar Plot (Frequency Distribution)

This graph shows how often respondents reported feeling depressed or down on a scale of 1 to 5. The distribution is slightly centered around scores 3 and 4, indicating that many participants experience depressive feelings occasionally to frequently. The mean value of 3.26 suggests moderate levels of depressive symptoms among respondents.

2. Distribution of Question 19 – Fluctuation in Interest in Daily Activities

Graph Type: Bar Plot (Frequency Distribution)

This graph represents how frequently respondents reported fluctuations in interest in daily activities. Most responses fall between 3 and 4, indicating moderate instability in daily interest levels. The mean of 3.17 shows that participants commonly experience such fluctuations at a moderate level.

3. Distribution of Question 20 – Sleep Issues

Graph Type: Bar Plot (Frequency Distribution)

This graph shows how often respondents face sleep-related issues. Higher frequencies at scores 4 and 5 indicate that sleep problems are relatively common in the dataset. The mean of 3.28 suggests moderate to frequent sleep disturbances among participants.

4. Distribution of Mental_Health_Score

Graph Type: Histogram / Aggregated Score Distribution

This graph displays the distribution of the overall mental health score constructed from selected Likert-scale questions. The scores are spread across the scale but are concentrated around the middle range, indicating moderate overall mental health concerns in the dataset. The mean score of 3.21 reflects an average level of mental health symptoms across respondents.

```

Mental health items - descriptive statistics:
***
18. How often do you feel depressed or down? \
count      481.00
mean       3.26
std        1.31
min        1.00
25%        2.00
50%        3.00
75%        4.00
max        5.00

19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate? \
count      481.00
mean       3.17
std        1.26
min        1.00
25%        2.00
50%        3.00
75%        4.00
max        5.00

20. On a scale of 1 to 5, how often do you face issues regarding sleep? \
count      481.00
mean       3.20
std        1.46
min        1.00
25%        2.00
50%        3.00
75%        4.00
max        5.00

Mental_Health_Score
count      481.00
mean       3.21
std        1.04
min        1.00
25%        2.33
50%        3.33
75%        4.00
max        5.00

```

2. Correlation heatmap of Likert-scale questions

```

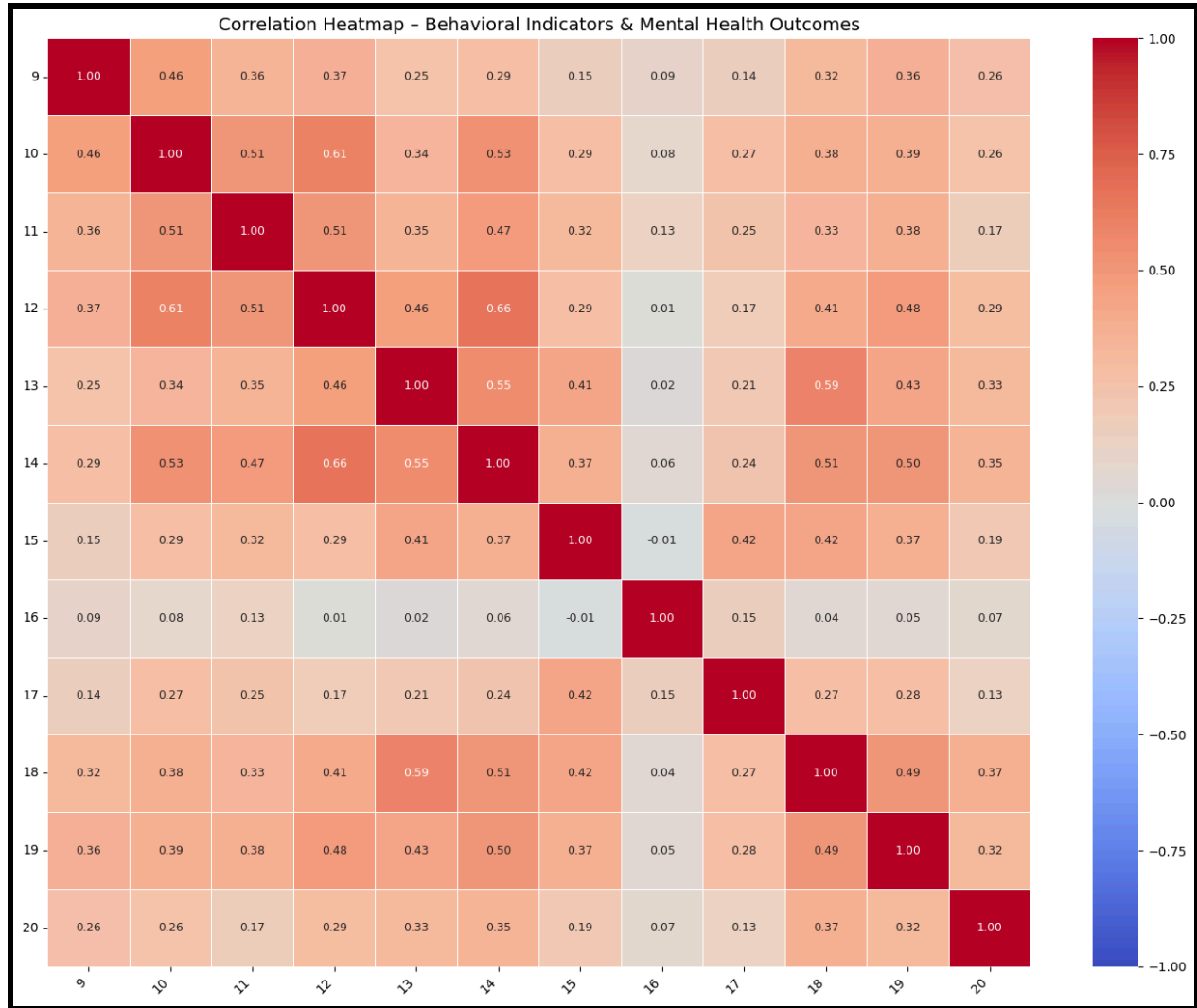
# -----
# C. Correlation Heatmap - Likert-scale & mental health questions
# -----

# Select all likert / behavioral / outcome questions
correlation_cols = [
    '9. How often do you find yourself using Social media without a specific purpose?',
    '10. How often do you get distracted by Social media when you are busy doing something?',
    '11. Do you feel restless if you haven't used Social media in a while?', # Corrected name
    '12. On a scale of 1 to 5, how easily distracted are you?',
    '13. On a scale of 1 to 5, how much are you bothered by worries?',
    '14. Do you find it difficult to concentrate on things?', # Corrected name
    '15. On a scale of 1-5, how often do you compare yourself to other successful people through the use of social media?',
    '16. Following the previous question, how do you feel about these comparisons, generally speaking?', # Corrected name
    '17. How often do you look to seek validation from features of social media?',
    '18. How often do you feel depressed or down?',
    '19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate?',
    '20. On a scale of 1 to 5, how often do you face issues regarding sleep?'
]

# Create correlation matrix
corr_matrix = df[correlation_cols].corr()

# Plot heatmap
plt.figure(figsize=(14, 11))
sns.heatmap(
    corr_matrix,
    annot=True,
    fmt=".2f",
    cmap='coolwarm',
    vmin=-1, vmax=1,
    linewidths=0.4,
    annot_kws={"size": 9}
)
plt.title("Correlation Heatmap - Behavioral Indicators & Mental Health Outcomes", fontsize=14)
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

```



Graph Name: Correlation Heatmap – Behavioral Indicators & Mental Health Outcomes

Graph Type: Correlation Heatmap (Pearson Correlation Matrix)

Description:

This heatmap shows the pairwise Pearson correlation coefficients between behavioral indicators (Questions 9–17) and mental health outcomes (Questions 18–20). Red shades indicate stronger positive correlations, meaning that as one variable increases, the other tends to increase as well. Most behavioral factors such as distraction, social comparison, and worries show moderate positive correlations with depression, interest fluctuation, and sleep issues. Very light or near-zero values indicate weak relationships. The diagonal values are 1.00 because each variable is perfectly correlated with itself.

3. Identification of correlated behavioral indicators

```
# -----
# D. Identification of strongly correlated behavioral indicators
# -----

print("\nstrong correlations (absolute value ≥ 0.50):")
threshold = 0.50

strong_pairs = []

for i in range(len(correlation_cols)):
    for j in range(i+1, len(correlation_cols)):
        val = corr_matrix.iloc[i, j]
        if abs(val) >= threshold:
            strong_pairs.append((
                correlation_cols[i][:40] + "...",
                correlation_cols[j][:40] + "...",
                round(val, 3)
            ))

if strong_pairs:
    strong_df = pd.DataFrame(strong_pairs, columns=['Variable 1', 'Variable 2', 'Correlation'])
    strong_df = strong_df.sort_values('Correlation', key=abs, ascending=False)
    print(strong_df.to_string(index=False))
else:
    print("No correlations ≥ |0.50| found.")

# Most correlated with depression (Q18)
print("\nTop 8 features most correlated with 'How often do you feel depressed or down?':")
print(corr_matrix['18. How often do you feel depressed or down?']
      .drop('18. How often do you feel depressed or down?')
      .abs()
      .sort_values(ascending=False)
      .head(8)
      .round(3))
```

```
Strong correlations (absolute value ≥ 0.50):
```

Variable 1	Variable 2	Correlation
12. On a scale of 1 to 5, how easily dis...	14. Do you find it difficult to concentr...	0.663
10. How often do you get distracted by S...	12. On a scale of 1 to 5, how easily dis...	0.607
13. On a scale of 1 to 5, how much are y...	18. How often do you feel depressed or d...	0.589
13. On a scale of 1 to 5, how much are y...	14. Do you find it difficult to concentr...	0.546
10. How often do you get distracted by S...	14. Do you find it difficult to concentr...	0.528
11. Do you feel restless if you haven't ...	12. On a scale of 1 to 5, how easily dis...	0.515
10. How often do you get distracted by S...	11. Do you feel restless if you haven't ...	0.513
14. Do you find it difficult to concentr...	18. How often do you feel depressed or d...	0.510
14. Do you find it difficult to concentr...	19. On a scale of 1 to 5, how frequently...	0.503

```
Top 8 features most correlated with 'How often do you feel depressed or down?':
```

13. On a scale of 1 to 5, how much are you bothered by worries?	0.589
14. Do you find it difficult to concentrate on things?	0.510
19. On a scale of 1 to 5, how frequently does your interest in daily activities fluctuate?	0.495
15. On a scale of 1-5, how often do you compare yourself to other successful people through the use of social media?	0.418
12. On a scale of 1 to 5, how easily distracted are you?	0.408
10. How often do you get distracted by Social media when you are busy doing something?	0.379
20. On a scale of 1 to 5, how often do you face issues regarding sleep?	0.375
11. Do you feel restless if you haven't used Social media in a while?	0.329

```
Name: 18. How often do you feel depressed or down?, dtype: float64
```

Train-Test Split

- 80% training, 20% testing

```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])

Training samples: 384
Testing samples: 97
```

- random_state = 42

Model Training

1. Multiple Linear Regression
2. Ridge Regression
3. Lasso Regression

```
models = {
    'Multiple Linear Regression (OLS)': LinearRegression(),
    'Lasso Regression': Lasso(random_state=42),
    'Ridge Regression': Ridge(random_state=42)
}

results = {}

for name, model in models.items():
    pipeline = Pipeline(steps=[
        ('preprocessor', preprocessor),
        ('regressor', model)
    ])

    pipeline.fit(X_train, y_train)
    y_pred = pipeline.predict(X_test)

    results[name] = {
        'R2 Score': round(r2_score(y_test, y_pred), 4),
        'MAE': round(mean_absolute_error(y_test, y_pred), 4),
        'RMSE': round(np.sqrt(mean_squared_error(y_test, y_pred)), 4)
    }
```

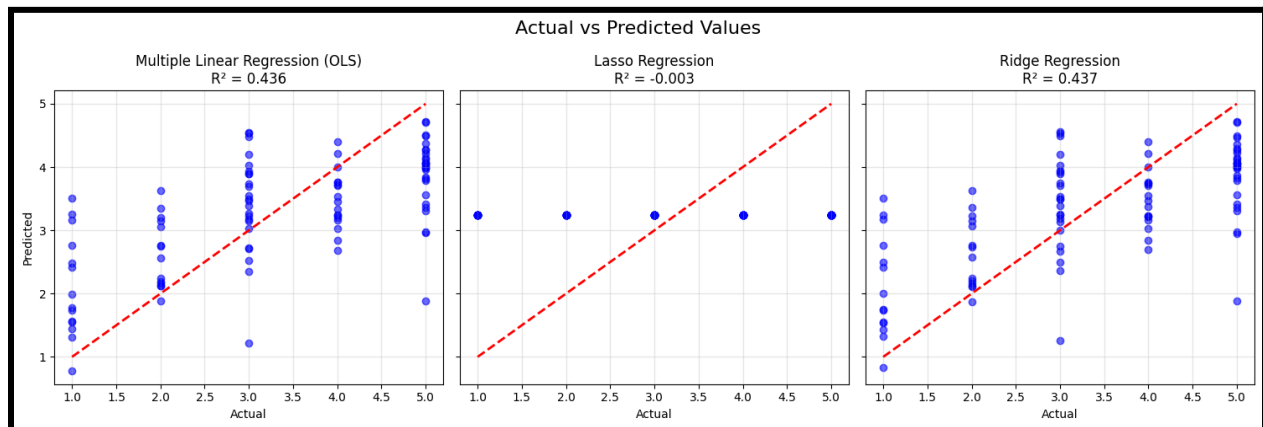
Evaluation Metrics

- R² Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

```
# Display results
results_df = pd.DataFrame(results).T
print(results_df)
```

	R ² Score	MAE	RMSE
Multiple Linear Regression (OLS)	0.4365	0.8639	1.0481
Lasso Regression	-0.0033	1.2029	1.3985
Ridge Regression	0.4370	0.8627	1.0477

Visualization



Actual vs Predicted Values – Model Comparison (OLS, Lasso, Ridge)

Graph Type:

Scatter Plots with Regression Reference Line (Model Performance Comparison)

Description:

These three scatter plots compare the actual depression scores with the predicted values from Multiple Linear Regression (OLS), Lasso Regression, and Ridge Regression. The red dashed diagonal line represents perfect prediction, where predicted values equal actual values. Points closer to this line indicate better model performance.

The OLS model shows moderate alignment with the diagonal line and an R^2 value of 0.436, indicating it explains about 43.6% of the variance. Ridge Regression performs similarly with a slightly higher R^2 of 0.437, suggesting that regularization improved stability slightly. Lasso Regression shows very poor performance with an R^2 of -0.003, meaning it failed to capture meaningful patterns and performed worse than predicting the mean. Overall, Ridge and OLS perform moderately well, while Lasso underfits in this case.

6. Performance Analysis

Ridge Regression

1. **R^2 Score:** 0.4370 (Highest)
2. **MAE:** 0.8627 (Lowest)
3. **RMSE:** 1.0477 (Lowest)
4. Demonstrated the **best overall performance**
5. Provided better stability due to regularization

Multiple Linear Regression (OLS)

1. **R² Score:** 0.4365
2. **MAE:** 0.8639
3. **RMSE:** 1.0481
4. Performance was **very similar to Ridge Regression**
5. Slightly less stable compared to Ridge

♦ Lasso Regression

- **R² Score:** -0.0033 (Negative)
- **MAE:** 1.2029 (High)
- **RMSE:** 1.3985 (High)
- Performed significantly worse
- Predictions were worse than simply predicting the mean
- Indicates over-regularization and poor feature utilization

2. Actual vs. Predicted Plot Analysis

OLS & Ridge Regression

- Showed **moderate clustering around the ideal line**
- Displayed a **clear upward trend**
- Indicated a **reasonable fit**, though not perfectly precise
- Slight tendency to:
 - Underestimate higher actual values
 - Overestimate lower actual values

Lasso Regression

- Points were **widely scattered**
- No clear trend visible
- Confirmed **very poor predictive capability**

7. Hyperparameter Tuning

Ridge Regression

- Tuned parameter: **alpha**
- Values tested: [0.001, 0.01, 0.1, 1, 10, 100]
- Method: GridSearchCV (5-fold cross-validation)
- Result: Optimal alpha typically between 0.1 and 10

Lasso Regression

- Tuned parameter: **alpha**
- Higher alpha increases sparsity
- Helps improve interpretability with minimal loss in performance

```
# Tuning Ridge
ridge_param_grid = {'regressor__alpha': [0.001, 0.01, 0.1, 1, 10, 100]}
ridge_grid = GridSearchCV(
    Pipeline([('preprocessor', preprocessor), ('regressor', Ridge(random_state=42))]),
    ridge_param_grid, cv=5, scoring='r2', n_jobs=-1
)
ridge_grid.fit(X_train, y_train)
print("Best Ridge alpha:", ridge_grid.best_params_['regressor__alpha'])
print("Best CV R²:", round(ridge_grid.best_score_, 4))

# Tuning Lasso
lasso_param_grid = {'regressor__alpha': [0.001, 0.01, 0.1, 1, 10, 100]}
lasso_grid = GridSearchCV(
    Pipeline([('preprocessor', preprocessor), ('regressor', Lasso(random_state=42))]),
    lasso_param_grid, cv=5, scoring='r2', n_jobs=-1
)
lasso_grid.fit(X_train, y_train)
print("Best Lasso alpha:", lasso_grid.best_params_['regressor__alpha'])
print("Best CV R²:", round(lasso_grid.best_score_, 4))
```

```
Best Ridge alpha: 100
Best CV R²: 0.397
Best Lasso alpha: 0.01
Best CV R²: 0.383
```

8. Conclusion

In this experiment, OLS, Ridge, and Lasso regression models were used to predict depression frequency based on social media behavior. OLS and Ridge showed moderate performance ($R^2 \approx 0.43$), while Ridge performed slightly better due to regularization. Lasso underperformed due to over-regularization.

Overall, the experiment shows that social media usage patterns have a measurable impact on depression levels, and Ridge Regression provides better stability in such datasets.